

Computer Vision:

Week-4 (09-13 Feb) Theory Lectures

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Review

- Invariant to scale
- Invariant to rotation
- Scale invariant feature transform
- Speeded up robust features
- Feature detection and descriptor

Introduction to Feature Matching

In computer vision, **feature matching** is the process of finding corresponding features between two images. After detecting keypoints and extracting descriptors (such as SIFT, SURF, ORB), matching is performed to identify similar descriptors across images.

- *Object recognition*
- *Image stitching (panorama)*

Feature Descriptors Recap

A feature descriptor is a numerical vector that represents the local appearance around a keypoint.

Examples: SIFT: 128-dimensional vector (floating point)

SURF: 64-dimensional vector

ORB: Binary descriptor

Matching compares these vectors to find the most similar ones.

Brute-Force (BF) Matching

- **Brute-Force matching** compares each descriptor from image 1 with all descriptors from image 2 and selects the closest match based on a distance metric.
- It is simple but computationally expensive.

Distance Metrics Used

- Euclidean (L2) distance → SIFT, SURF
- Hamming distance → ORB, BRIEF

Distance formula (Euclidean):

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

Advantages and Disadvantages

Advantages: Very simple to understand and implement

Accurate for small datasets

Disadvantages: Slow for large number of features

Time complexity is high

Euclidean distance

Suppose a descriptor from Image A is :

$$A = [2, 4]$$

Descriptors from Image B :

$$B1 = [3, 5]$$

$$B2 = [10, 12]$$

Euclidean distance

Compute Euclidean distance:

$$D(A, B_1) = \sqrt{(2-3)^2 + (4-5)^2} = \sqrt{2} \approx 1.41$$

$$D(A, B_2) = \sqrt{(2-10)^2 + (4-12)^2} = \sqrt{128} \approx 11.31$$

Best match = B_1 (minimum distance)

FLANN-Based Matching

What is FLANN?

- FLANN stands for Fast Library for Approximate Nearest Neighbors.
- Instead of checking all descriptors, FLANN uses efficient search structures (like KD-trees) to find nearest neighbors quickly.

Why FLANN is Faster

- Uses approximate nearest neighbor search
- Reduces computation time
- Suitable for large datasets

Comparison

<i>Feature</i>	<i>Brute-Force</i>	<i>FLANN</i>
<i>Speed</i>	Slow	Fast
<i>Accuracy</i>	Exact	Approximate
<i>Best for</i>	Small datasets	Large datasets
<i>Complexity</i>	High	Lower

The Ratio Test

- Sometimes the closest match is not reliable.
- The ratio test checks whether the best match is significantly better than the second-best match.

Ratio Test

Best match distance = 40 Second – best match distance = 70

Ratio = $40 / 70 = 0.57$

Since $0.57 < 0.75 \rightarrow$ Good match accepted

Ratio Test

Another case :

$$\textit{Best} = 40 \textit{ Second} - \textit{best} = 45$$

$$\textit{Ratio} = 40 / 45 = 0.88$$

Since $0.88 > 0.75 \rightarrow$ **Match rejected**

Comaprison

Combining FLANN with Ratio Test

In practice:

1. Use **FLANN** to get nearest neighbors
2. Apply **ratio test** to remove false matches

This improves both speed and accuracy.

Numerical

**Euclidean Distance (2-D
descriptor)**

Given:

Descriptor A = [1, 2]

Descriptor B = [4, 6]

Given: (3-D)

Descriptor A = [2, 3, 5]

Descriptor B = [4, 1, 6]

Numerical-Ratio Test

Given:

Best match distance = 30

Second-best match distance = 50

Threshold = 0.75

Computation Cost – BF vs FLANN

Given:

Image A has 500 descriptors

Image B has 500 descriptors

Computation Cost – BF vs FLANN

(a) Brute-Force Matching

BF compares every descriptor with all others:

Comparisons = $500 \times 500 = 250,000$

Computation Cost – BF vs FLANN

(b) FLANN Matching

FLANN uses approximate search and reduces comparisons.

Assume FLANN checks only 10% of
comparisons: $250,000 \times 0.1 = 25,000$